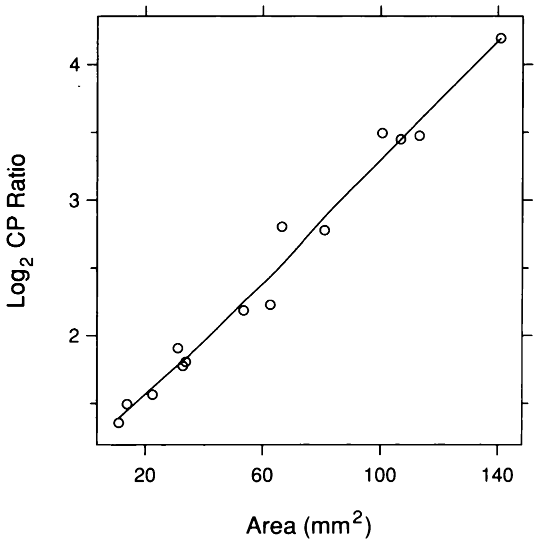
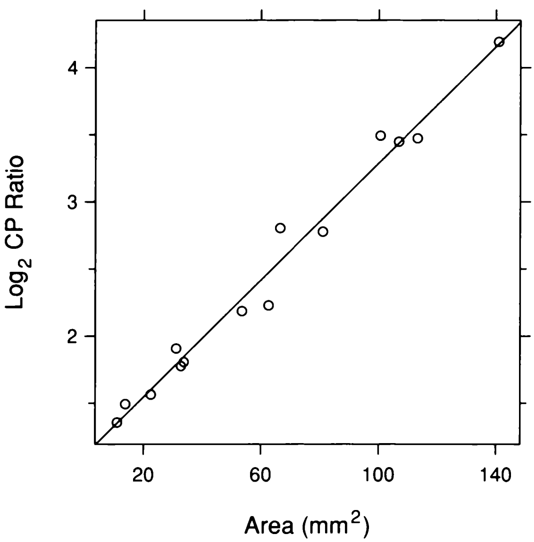
As with univariate data, transformation — in this case, of the response — can cure monotone spread. Figure 3.15 graphs log CP ratio against area with a loess curve superposed. Something fortuitous has occurred; the overall pattern now appears nearly linear, which is simpler than the curved pattern before transformation. Figure 3.16 shows a line

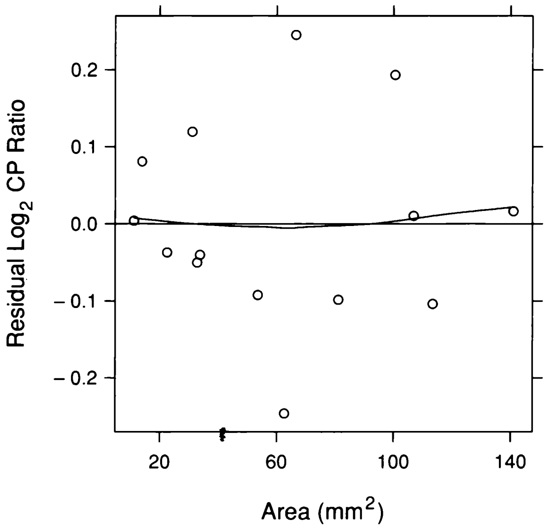



3.15 Log CP ratio is graphed against area. The curve is a loess fit with $\alpha = 2/3$ and $\lambda = 1$.

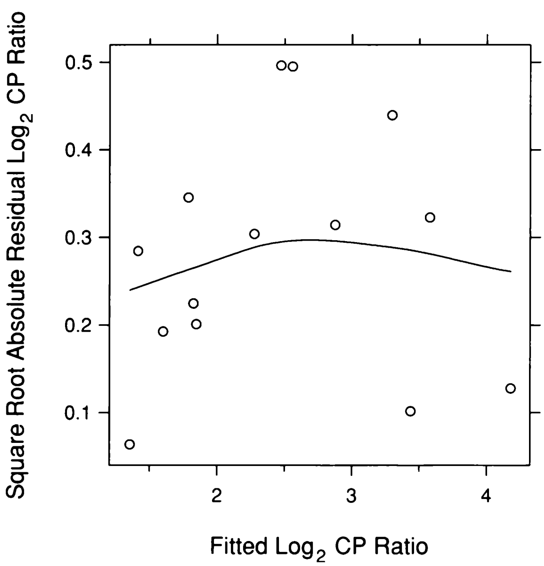



3.16 Log CP ratio is graphed against area. The line is the least-squares fit.

fitted by least-squares, and Figure 3.17 is a residual dependence plot. There is no dependence of the location of the residuals on area. Figure 3.18 is an s-l plot. There is no longer any dependence of spread on location; the log transformation has removed the monotone spread and, along the way, the nonlinearity of the underlying pattern.



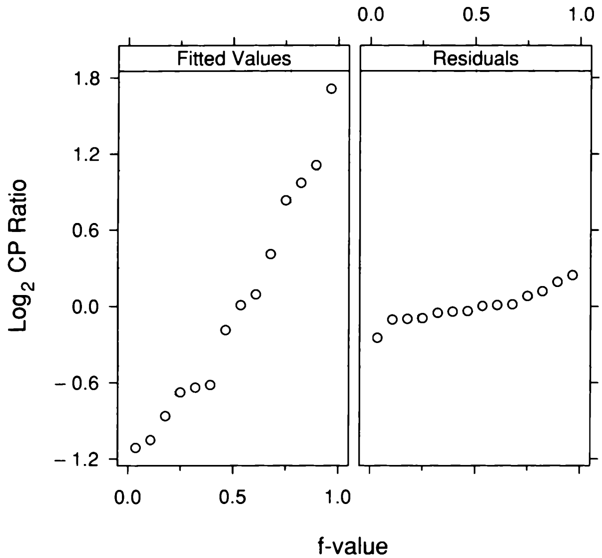

3.17 On this residual dependence plot, the residuals from the linear fit to log CP ratio are graphed against area. The parameters of the loess curve on the plot are $\alpha = 1$ and $\lambda = 1$.




3.18 An s-l plot for the linear fit to log CP ratio checks for monotone spread. The parameters of the loess curve are $\alpha = 2$ and $\lambda = 1$.

*R-F Spread Plots*

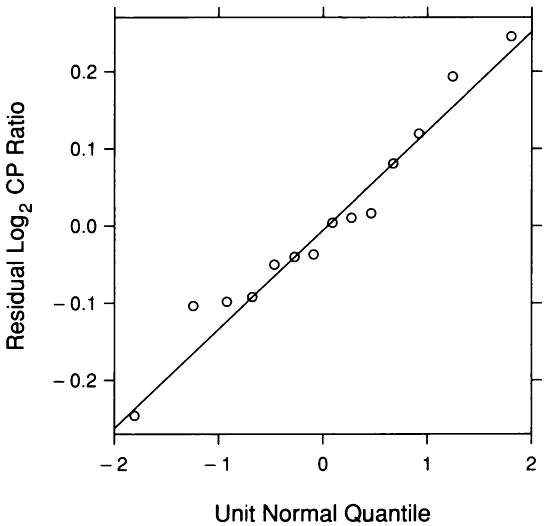
As with univariate data, an r-f spread plot consists of quantile plots that compare the spreads of the residuals and the fitted values minus their mean. This shows how much variation in the data is explained by the fit and how much remains in the residuals. Figure 3.19 is an r-f spread plot for the fit to log CP ratio. The spread of the residual distribution is considerably smaller than that of the fitted values; the fit explains most of the variation in the data.




3.19 An r-f spread plot compares the spreads of the residuals and the fitted values minus their mean for the linear fit to log CP ratio.

*Fitting the Normal Distribution*

Figure 3.20, a normal quantile plot of the residuals, shows that the residual distribution is well approximated by the normal. This justifies our use of least-squares to fit the data. The theory of statistics tells us that least-squares is appropriate for bivariate fitting when the distribution of the random variation of the response about the underlying pattern is well approximated by the normal, just as it tells us that means are appropriate for univariate fitting when the variation of univariate data is well approximated by the normal. When outliers or other forms of leptokurtosis are present in the residuals, robust methods of fitting are needed, just as they are for univariate data. This is taken up in Section 3.4.




3.20 A normal q-q plot compares the normal distribution with the distribution of the residuals for the linear fit to log CP ratio.

*The Dependence of Ratio on Area*

The final result of the visualization of the ganglion data is a simple description of how CP ratio depends on area. The simplicity arises because log CP ratio is linear in area, the residuals are homogeneous, and the normal distribution provides a good approximation to the residual distribution. Let $y$ be the $\log_2$ ratio and let $x$ be area. The least-squares fit is

$$y = 1.11 + 0.022x \,.$$

If area changes by $1/0.022\ \text{mm}^2 = 45\ \text{mm}^2$, the $\log_2$ CP ratio changes by 1, which means the CP ratio doubles. The areas of the ganglion data set vary by $130\ \text{mm}^2$, so $\log_2$ CP ratio changes by about 2.8 overall, and the CP ratio changes by a factor of $2^{2.8} = 7.0$. The sample standard deviation of the residuals is 0.13. Thus, using the normal approximation to the residuals, about 95% of the residual distribution lies between $\pm 1.96 \times 0.13 = \pm 0.25$, which on the CP ratio scale are factors of 0.84 and 1.19.

*Log Transformation*

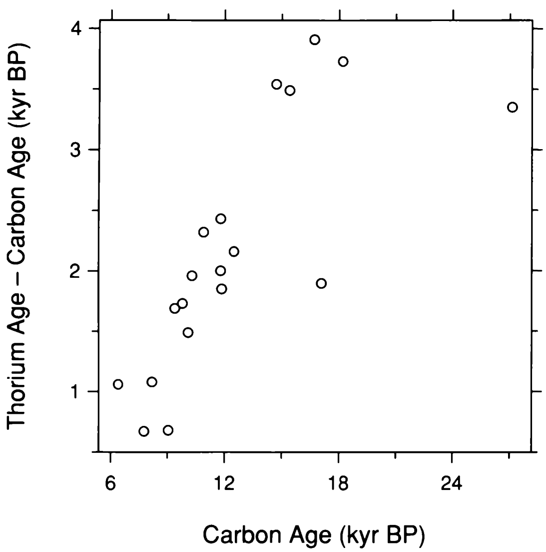
The tortuous path to the logarithm in the analysis of the ganglion data might have been avoided at the outset. The scatterplot of the data with the loess fit in Figure 3.2, which showed a convex pattern, might have inspired us to take logs to attempt a straightening of the pattern, rather than moving to a quadratic fit. But even more fundamentally, we might have taken the logs of the ratios before visualizing the data. A ratio, which has a numerator and denominator, is a strong candidate for a log transformation. When the numerator is bigger, the ratio can go from 1 all the way to infinity, but when the denominator is bigger, the ratio is squeezed into the interval 0 to 1. The logarithm symmetrizes the scale of measurement; when the numerator is bigger, the log ratio is positive, and when the denominator is bigger, the log ratio is negative. The symmetry often produces simpler structure.

## 3.4 Robust Fitting

Ages of many ancient objects, such as archeological artifacts and fossils, are determined by carbon dating. Organic material absorbs $^{14}C$, which then decays at a known rate through time. Carbon dating, which consists of measuring the remaining $^{14}C$, is not a fully accurate timepiece because the amount of $^{14}C$ in the environment available for absorption by organic material has fluctuated through time. Tree-ring dating was at first used to calibrate carbon dating, but this method of calibration works only back to about 9 kyr BP (kiloyears before present).

A second dating method, first reported in 1990, provides calibration back to at least 30 kyr BP by measuring the decay of uranium to thorium [5]. The group that invented the method took core samples in coral off the coast of Barbados and dated the material back to nearly 30 kyr BP using both the carbon and thorium methods. The thorium results were used to study the accuracy of the carbon method. Of course, this is valid only if the thorium method is itself accurate, but the consensus appears to be that it is [64].

Figure 3.21 shows the results of dating 19 coral samples by both methods; thorium age minus carbon age is graphed against carbon age. The display shows that carbon dating using the old method of calibration underestimates the true age. The scatterplot suggests a linear pattern, but there are two outliers, deviant observations that do not conform to the pattern of the remaining observations. Both lie well below the linear pattern suggested by the other observations.




3.21 The differences of carbon age and thorium age for 19 coral samples are graphed against the carbon ages.

Figure 3.22 shows the least-squares fit of a line to the dating data. The fit does not follow the underlying linear pattern; the slope is too small because the line has been pulled down by the outliers. Figure 3.23 is a residual dependence plot; there is a pronounced remaining dependence on carbon age.

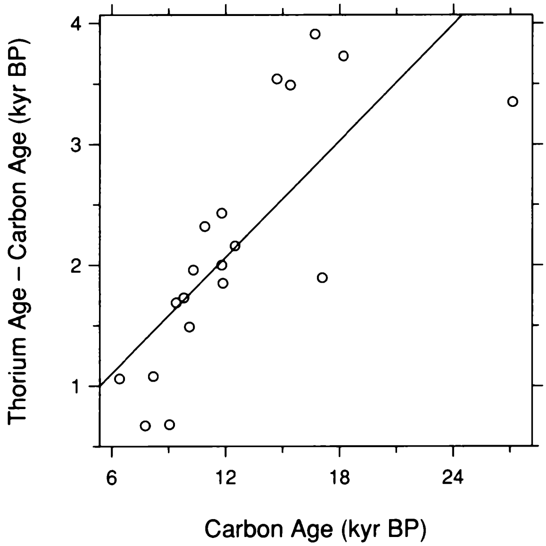
In Chapter 2, the distortion of the mean by outliers was discussed. Like the mean, least-squares estimates are not robust. If fact, the mean of univariate data is a least-squares estimate; it is the value of $a$ that minimizes

$$\sum_{i=1}^{n} (y_i - a)^2 .$$

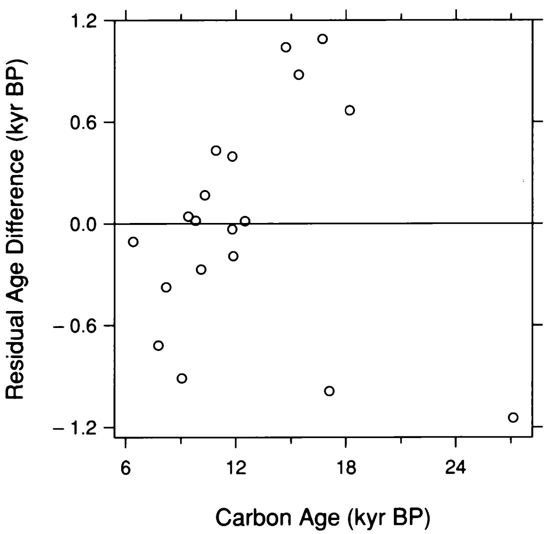
Least-squares is appropriate for bivariate fitting when the distribution of the random variation of the response about the underlying pattern is well approximated by the normal. And its performance is satisfactory if the distribution is moderately platykurtic. But in the presence of leptokurtosis, even in the form of just a few outliers, least-squares can be a disaster. And because loess is based on least-squares fitting of many polynomials locally, it too is not robust.

### *Bisquare*

In Chapter 2, when a distribution appeared leptokurtic, we moved to the median, a robust estimate of location that is not adversely affected by leptokurtosis. For bivariate data, we will use the robust estimation method *bisquare* [51, 65]. Bisquare is added to the least-squares method for fitting a parametric family and is added to loess to turn these nonrobust methods into robust ones.




3.22 The least-squares fit of a line has been added to the scatterplot of the dating data.




3.23 The residuals from the least-squares fit to the dating data are graphed against carbon age.

Figure 3.24 illustrates bisquare fitting of a line with the dating data. The first step is to fit by least-squares; this is shown in the upper right panel. The two outliers are graphed by the "+" plotting symbols. The second step is to assign a robustness weight, $r_i$, to each observation $(x_i, y_i)$, based on the size of the residual for that observation. The weights are assigned by the value of a weight function called *bisquare*, from which the fitting procedure gets its name. The residuals and the weight function are graphed in the middle left panel of Figure 3.24. Residuals close to 0 have a weight close to 1. As the absolute values of the residuals increase, the weights decrease; for example, the largest and smallest residuals have weights of about 0.6. In the next step, the parameters are re-estimated using weighted least-squares with the weights $r_i$. In our example, we are fitting the parametric family

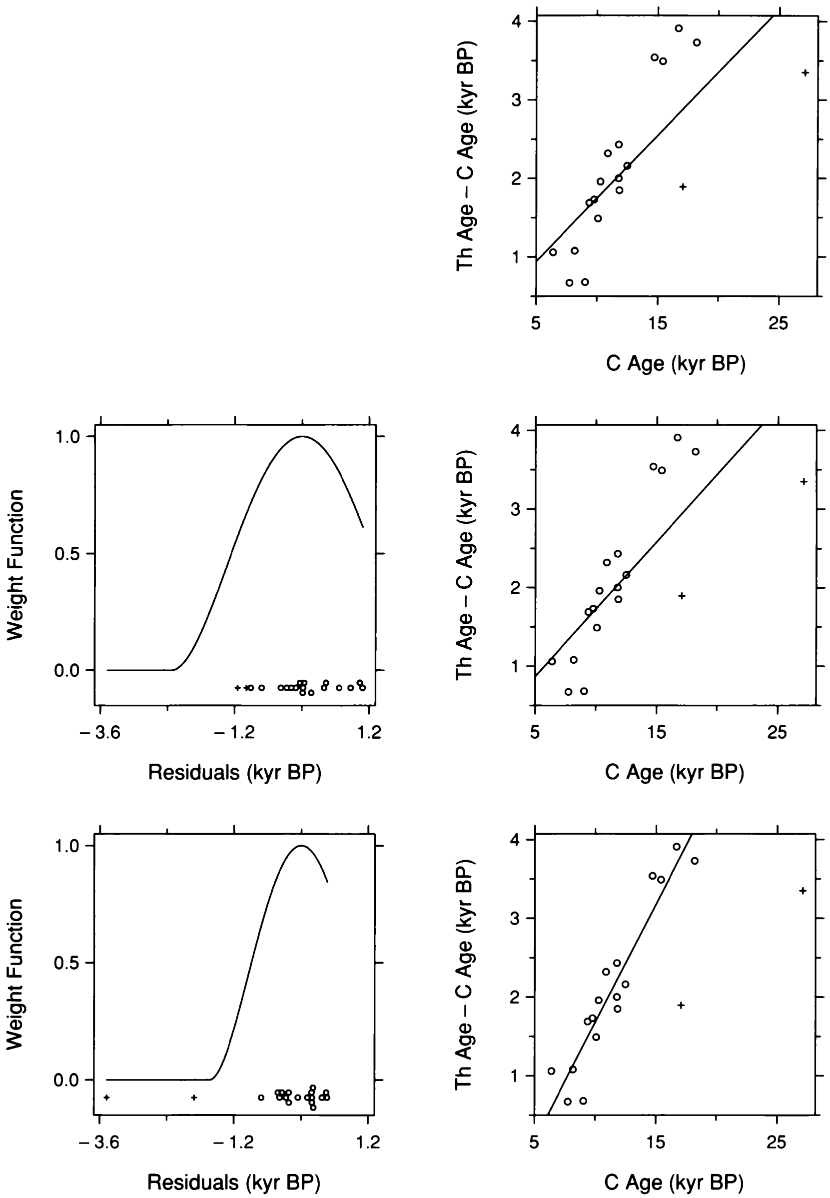
$$y = a + bx \; .$$

Thus, the weighted least-squares estimates are the values of $a$ and $b$ that minimize

$$\sum_{i=1}^{n} r_i(y_i - a - bx_i)^2 \; .$$

The new fit is shown in the middle right panel of Figure 3.24. Next, new weights are computed, and a line is fitted using weighted least-squares with the new weights. This iterative procedure is continued until the fit converges. The bottom left panel of Figure 3.24 shows the weights for the final fit, and the bottom right panel shows the fit; the outliers receive zero weight in this final fit.

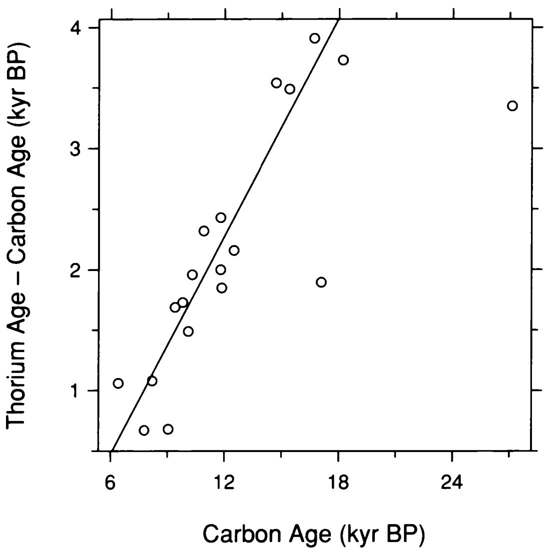
In a similar way, loess can be altered by the bisquare method to make it robust. The details of how bisquare iterations are added to least-squares parametric fitting and how the iterations are added to loess curve fitting are given at the end of this section.



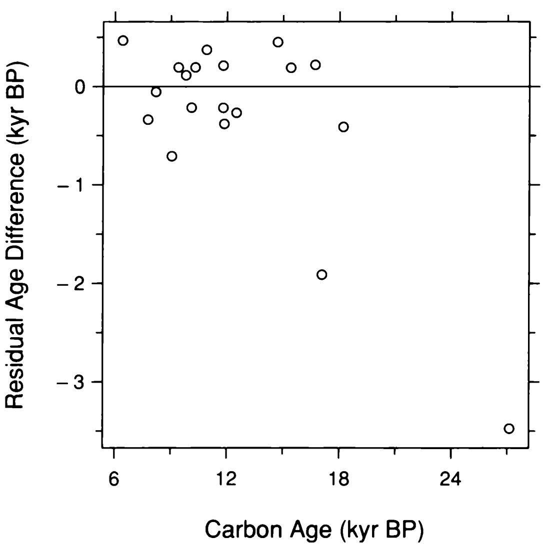

3.24 The figure illustrates the fitting of a line to the dating data by the bisquare method of robust estimation.

*A Robust Analysis of the Dating Data*

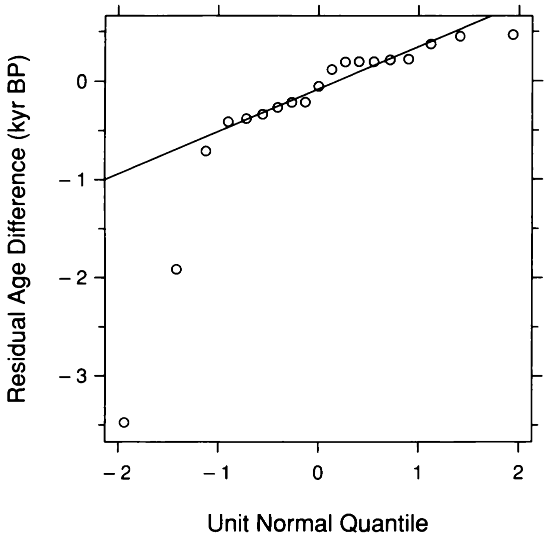
The bisquare line is graphed again in Figure 3.25. A residual dependence plot is shown in Figure 3.26. There is no remaining dependence in the residuals; the fit describes the overall pattern of the majority of the data. Figure 3.27 is a normal quantile plot of the residuals. Except for the outliers, the points have a linear pattern.




3.25 The bisquare fit of a line has been added to the scatterplot of the dating data.




3.26 The residuals from the bisquare fit are graphed against carbon age.




3.27 A normal q-q plot displays the residuals from the bisquare fit to the dating data.

The equation of the fitted bisquare line is

$$y = -1.3 + 0.30x \; .$$

The response $y$ is zero when $x$ is 4.3 kyr BP. Thus the fitted line suggests that a departure between carbon and thorium ages begins near 4.3 kyr BP. After that time point, the underestimation of age by carbon dating grows at a rate of 0.3 kyr for each kyr of carbon age. Thus for a carbon age of 15 kyr BP, the underestimation is about

$$0.3(15 - 4.3) \text{ kyr BP} = 3.2 \text{ kyr BP} \; .$$

The underestimation of age by traditional carbon dating is substantial. Carbon dating specialists had been aware that the method was biased, but the magnitude of the bias came as a surprise [64].

### *The Scope of Bisquare*

Bisquare has been introduced in this chapter for use with bivariate data. It will also be used in coming chapters because it extends in a straightforward way to trivariate, hypervariate, and multiway data. For univariate data, the median was used in Chapter 1 for robust estimation because it is simple, widely known, and almost always performs well when robustness to outliers is needed. But we can also use bisquare for univariate data; the initial estimate is the mean, and subsequent estimates in the iterations are weighted means with bisquare weights computed from the residuals [51, 65]. Thus bisquare is attractive because a single method can be used for all the data types in the book to produce robust estimates.

### *For the Record: The Details of Bisquare*

For fitting a parametric family, the only detail of bisquare that remains is the formula for the robustness weights, $r_i$. Let $s$ be the median absolute deviation for the least-squares fit or any subsequent weighted fit. Thus

$$s = \text{median}\,|\widehat{\varepsilon}_i|,$$

where the $\widehat{\varepsilon}_i$ are the residuals from the fit. Let $B(u)$ be the bisquare weight function,

$$B(u) = \begin{cases} (1-u^2)^2 & \text{for } |u| < 1 \\ 0 & \text{otherwise}. \end{cases}$$

The robustness weight for the observation $(x_i, y_i)$ is

$$r_i = B\left(\frac{\widehat{\varepsilon}_i}{6s}\right).$$

As discussed in Chapter 2, $s$ is a measure of the spread of the residuals. If a residual is small compared with $6s$, the robustness weight will be close to 1; if a residual is greater than $6s$, the weight is 0. Large residuals tend to be associated with outliers, so unusual observations are given reduced weight. The constant 6 has been chosen so that if the random variation of the response about the underlying pattern has a distribution that is well approximated by the normal, then bisquare does nearly as well as least-squares.

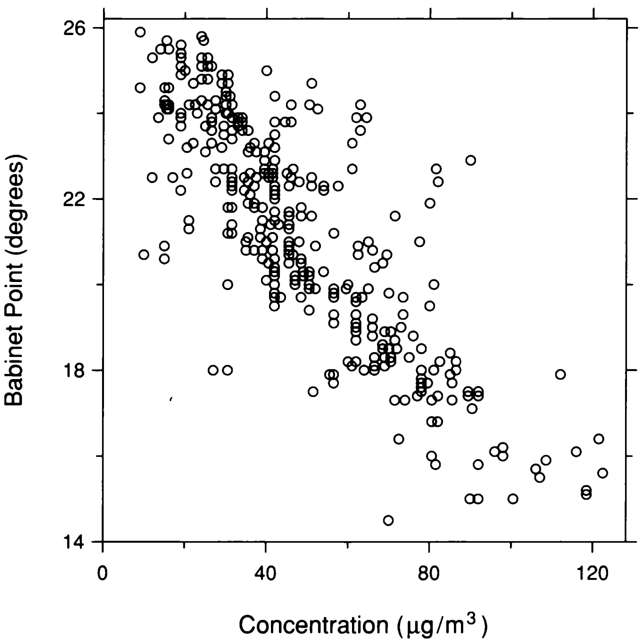
For loess, an initial fit is carried out with no robustness weights. The fit is evaluated at the $x_i$ to get the fitted values, $\widehat{y}_i$, and the residuals,

$$\widehat{\varepsilon}_i = y_i - \widehat{y}_i.$$

Bisquare weights, $r_i$, are computed from the residuals using the bisquare weight function exactly as described for fitting a parametric function. Thus each observation $(x_i, y_i)$ has a robustness weight, $r_i$, whose value depends on the magnitude of $\widehat{\varepsilon}_i$. Weighted local fitting is again carried out, but now the weight for $(x_i, y_i)$ for the fit at $x$ is $r_i w_i(x)$. Thus, $(x_i, y_i)$ is given reduced weight if $x_i$ is far from $x$ or if $\widehat{\varepsilon}_i$ is large in absolute value. The second fit is used to define new residuals which are used to define new robustness weights, and the local fitting is carried out again. This process is repeated until the loess curve converges.

## 3.5 *Transformation of a Factor and Jittering*

Figure 3.28 is a scatterplot of 355 observations from an experiment at the University of Seville on the scattering of sunlight in the atmosphere [7]. The response is the *Babinet point*, the scattering angle at which the polarization of sunlight vanishes. The factor is the concentration of particulates, which are solid particles in the air. These data were briefly discussed in Chapter 1.

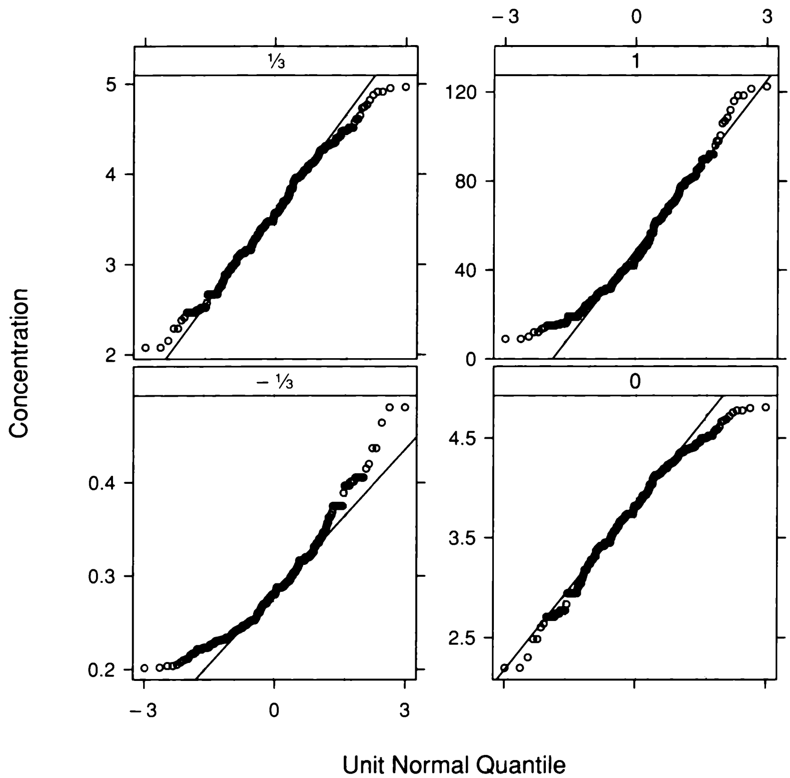



3.28 The scatterplot displays the polarization data.

*Transformation*

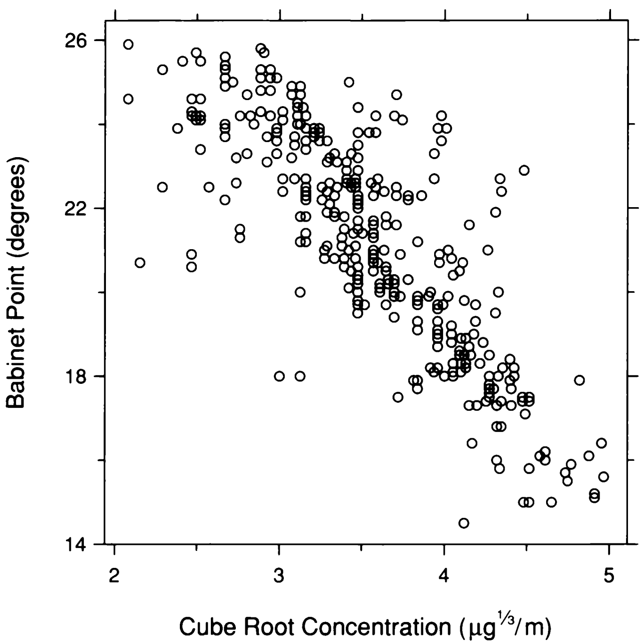
Transformation was first introduced in Chapter 2 as a tool for simplifying the structure of data. Power transformation cured skewness toward large values and monotone spread. For bivariate data, transformation of the response can also cure these ills. The ganglion data are one example.

For bivariate data, power transformation of the factor can also simplify structure. This is especially true of a factor whose measurements are observational and not specified by an experimenter. The particulate concentrations are one example; they are at the mercy of atmospheric conditions and the amount of air pollution.




3.29 Normal q-q plots display four power transformations of the particulate concentrations.

Figure 3.28 suggests that the particulate concentrations, as a univariate data set, are somewhat skewed toward large values. One result is a few observations in the lower right corner that straggle somewhat from the remaining. If a power transformation can remove the skewness, it might simplify the dependence of the Babinet point on concentration. Figure 3.29 shows normal q-q plots for four power transformations of the concentrations. The cube roots appear to symmetrize the distribution of the data. Figure 3.30 graphs the Babinet point against cube root concentration. The effect is not dramatic, but we have pulled the lower right stragglers into line.

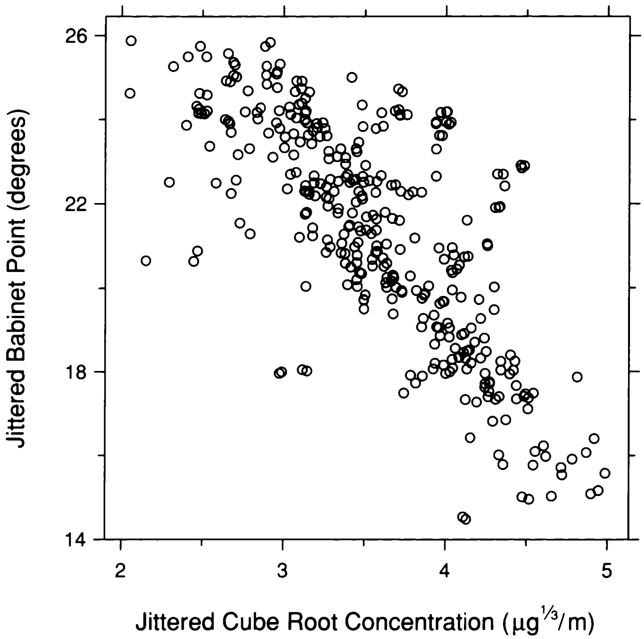



3.30 The measurements of the Babinet point are graphed against the cube root concentrations.

*Visual Perception: Jittering*

There is still a problem to be cured on the scatterplot in Figure 3.30; observations are obscured because the data on each scale are rounded, which results in a multiplicity of points at many of the plotting locations. One solution to the overlap problem, a particularly simple one, is jittering [20]. A small amount of uniform random noise is added to the data before graphing. It can be added to the measurements of just one of the variables or to the measurements of both.

Jittering is used in Figure 3.31 to display the polarization data. Noise has been added to the measurements of both variables; for each variable the noise was generated from a uniform distribution on an interval that is centered on zero and that is small compared with the range of the measurements. Now, most of the plotting symbols can be visually detected.




3.31 The measurements of the Babinet point are graphed again against the cube root concentrations, but this time the locations of the plotting symbols have been jittered.

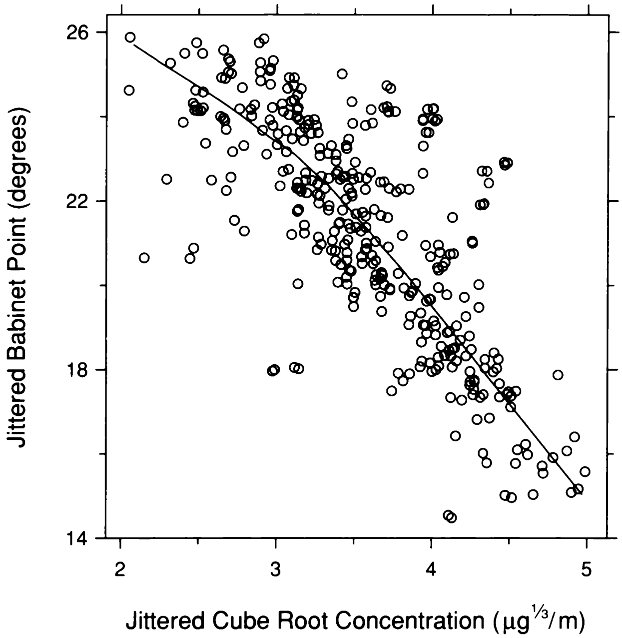
## 3.6 The Iterative Process of Fitting

Fitting mathematical structures to data must be relentlessly iterative. The analysis of the ganglion data provides one illustration. The first fit, a line fitted to CP ratio, was quite obviously inadequate. On the second pass, a loess fit showed a pattern that appeared simple enough to be fitted by a quadratic function. On the third pass, a quadratic polynomial was fitted; there was no lack of fit, but an s-l plot detected monotone spread. On the fourth pass, logs were taken and the loess fit showed a linear pattern. On the fifth pass, log CP ratio was fitted by a line. The fit passed the visualization tests for lack of fit, and the residuals proved to be homogeneous with a distribution well approximated by the normal.

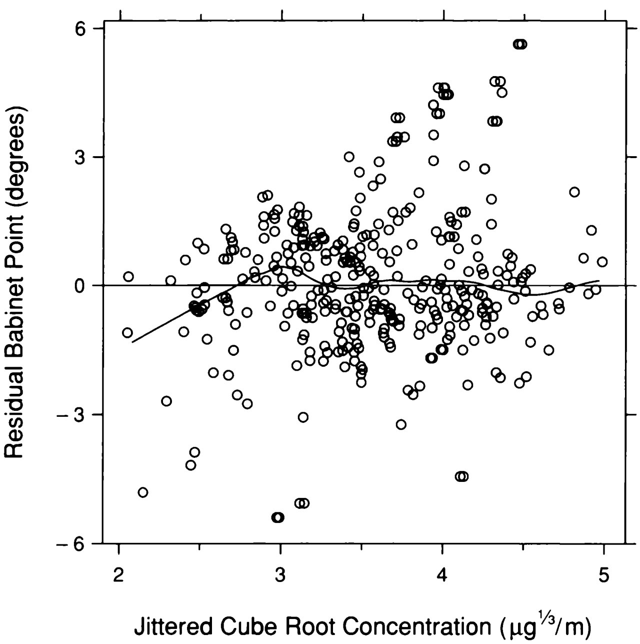
The single task of fitting a loess curve typically needs to be iterative as well. It is often possible to make an initial correct guess of $\lambda$ and of whether bisquare is needed or not, but the choice of $\alpha$ usually requires a few trial values, with the final choice based on the visualization of fits and residuals. The polarization data provide one example.

Interestingly, without a smooth curve superposed, Figure 3.31 appears to have an underlying pattern that is linear; loess fits will show that this is not the case. However, the scatterplot does indicate that $\lambda$ should be one, for even if we do not fully trust our unaided eyes, we can trust them enough to conclude that whatever curvature might be present is not likely to have peaks and valleys in need of locally quadratic fitting. Figure 3.31 also suggests that we should use bisquare iterations to make the loess fitting robust. There are a number of observations that deviate substantially in the vertical direction from the underlying pattern. For example, at a cube root concentration of $c = 3\sqrt[3]{\mu\text{g}}/\text{m}$, and a Babinet point of $b = 18^\circ$, there is a small cluster of deviant observations. Finally, we will use $\alpha = 3/4$ as a trial value.

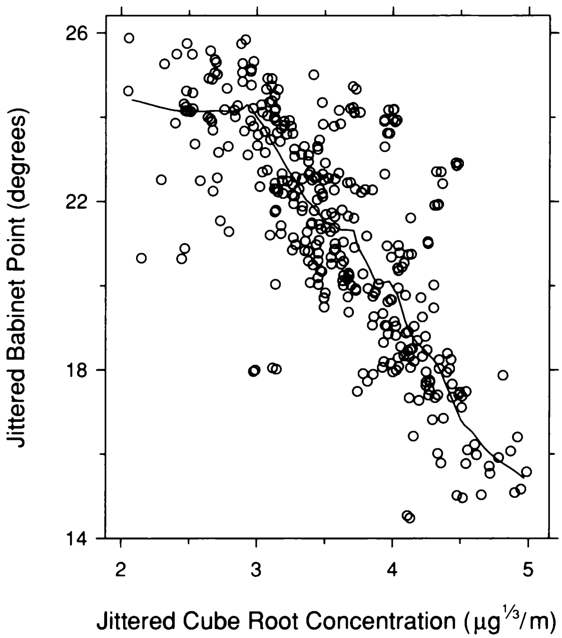
Figure 3.32 graphs a robust loess fit with $\alpha = 3/4$ and $\lambda = 1$. The pattern suggested by the curve is two straight lines with a break point near 3 $\sqrt[3]{\mu\text{g}}/\text{m}$. Figure 3.33 is a residual dependence plot. There appears to be a remaining dependence of the residuals on the factor. There is lack of fit because $\alpha$ is too large. Figure 3.34 graphs a robust loess fit with $\alpha$ reduced to 1/6, and Figure 3.35 is a residual dependence plot. There is no remaining residual dependence, but the fitted curve in Figure 3.34 has undue wiggles that do not appear to be supported by the data. There is surplus of fit because $\alpha$ is now too small. In choosing $\alpha$, we want a curve that is as smooth as possible without introducing lack of fit. Figure 3.36 graphs a robust loess fit with $\alpha$ increased to 1/3, and Figure 3.37 is a residual dependence plot. No residual dependence appears to be present, yet the fitted curve is smooth — the unnecessary wiggles are gone. These loess parameters appear to be a good choice. Figure 3.38 is a normal quantile plot of the residuals from the final fit. The residual distribution is strongly leptokurtic; the robust loess fitting is needed in this example. Figure 3.39 is an r-f spread plot. The amounts of variation in the two distributions are roughly commensurate. Most residuals lie between $\pm 3^\circ$, but a small fraction deviate by much more, some getting as large as $\pm 6^\circ$. The range of the fitted values is about $9^\circ$.



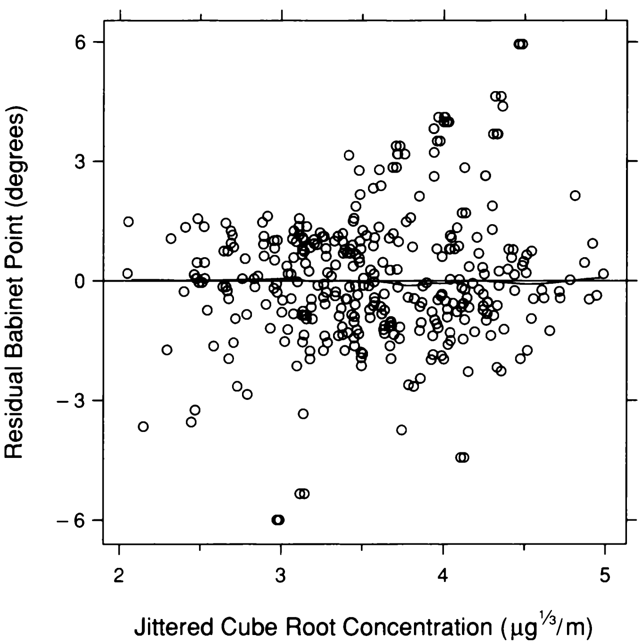

3.32 The curve is a robust loess fit to the polarization data with $\alpha = 3/4$ and $\lambda = 1$. The aspect ratio has been chosen to bank the curve to $45^\circ$.




3.33 The residuals from the robust loess fit are graphed against cube root concentration. The parameters of the robust loess curve on the plot are $\alpha = 1/3$ and $\lambda = 1$.




3.34 The curve is a robust loess fit to the polarization data with $\alpha = 1/6$ and $\lambda = 1$. The curve is banked to $45^{\circ}$.




3.35 The residuals from the second robust loess fit are graphed against cube root concentration. The parameters of the robust loess curve on the plot are $\alpha = 1/3$ and $\lambda = 1$.